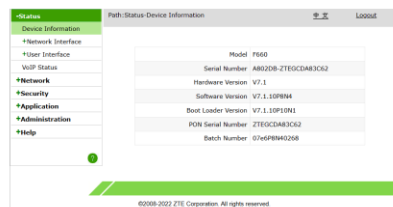
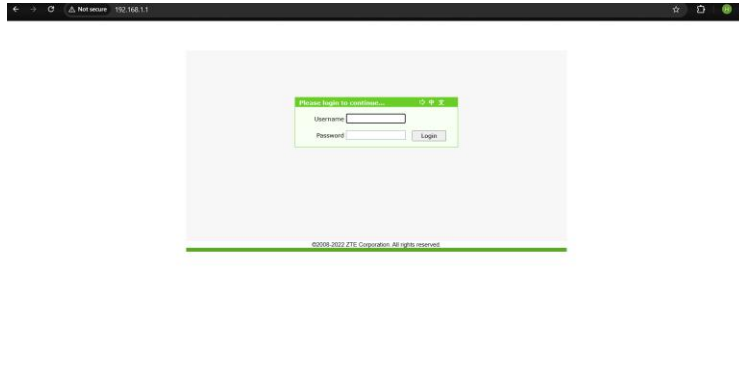


Configuring Basic Network HardwareTheory

1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page.



2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

After connecting the laptop to the Wi-Fi network, the network configuration was checked using the ipconfig command in Command Prompt.

Network Information Value

IPv4 Address 192.168.1.146

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

Result: The laptop is connected to the Wi-Fi network with the IP address **192.168.1.146**, subnet mask **255.255.255.0**, and default gateway **192.168.1.1**.

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is 192.168.1.1), and test if you still have internet access.

Auto Dhcp

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2401:4900:8fc6:de60:4a41:3ba8:ab8b:5610
Temporary IPv6 Address. . . . . : 2401:4900:8fc6:de60:1931:a5a:c44d:639c
Link-local IPv6 Address . . . . . : fe80::436e:d56e:4714:842c%15
IPv4 Address. . . . . : 192.168.1.146
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::26de:8aff:fe79:8d11%15
                            192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

C:\Users\Harsh>
```

Manual (Static)

```
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

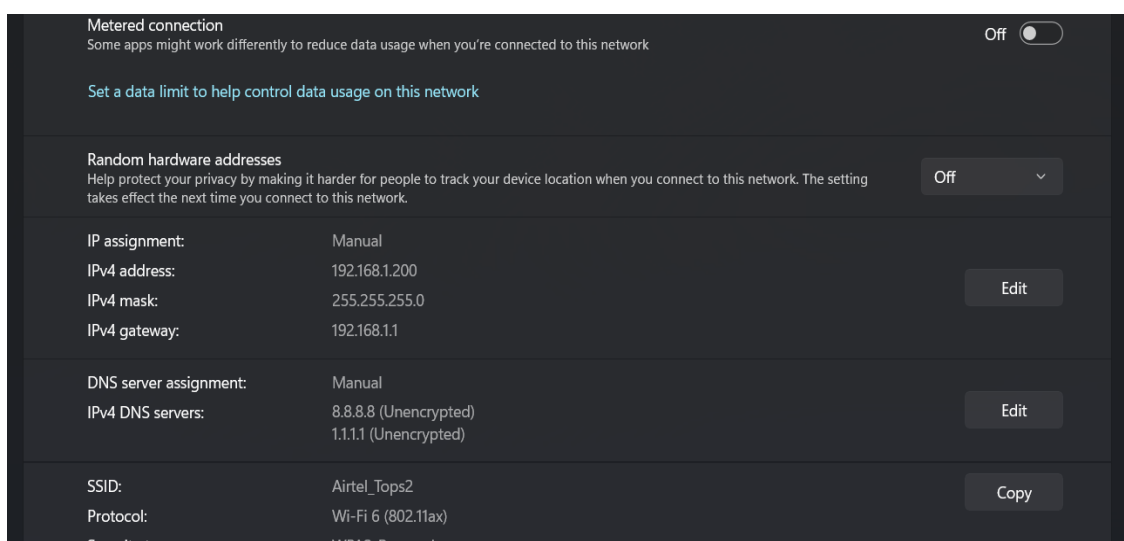
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. . . . . : 2401:4900:8fc6:de60:4a41:3ba8:ab8b:5610
Temporary IPv6 Address. . . . . : 2401:4900:8fc6:de60:1931:a5a:c44d:639c
Link-local IPv6 Address . . . . . : fe80::436e:d56e:4714:842c%15
IPv4 Address. . . . . : 192.168.1.200
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : fe80::26de:8aff:fe79:8d11%15
                            192.168.1.1

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :

C:\Users\Harsh>
```



4. In your router's admin panel, locate the section where you can view or change the DHCP

range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

DHCP Range

The DHCP range configured on the router is **192.168.1.100 to 192.168.1.200**. This range allows the router to automatically assign IP addresses to devices connected to the network.

If two devices are assigned the **same IP address**, an **IP address conflict** occurs. As a result, one or both devices may experience network connectivity problems and may not be able to communicate properly with other devices.

DHCP Range: 192.168.1.100 – 192.168.1.200